Capstone Project

Project Title: Agentic AI Assistant for a Retail Company

Project Type: Hands-on Capstone Project

Organization: Analytics Vidya

Learner Name: <ENTER HERE>

Project Description / Business Case:

The scenario is that you are working for a retail company, ShopUNow and the focus of this use case is to **build an Intelligent AI Assistant** which can leverage internal company information to answer both internal employee as well as external customer queries. This is a classic problem which could be solved leveraging **aspects of Agentic AI and RAG Systems**.

The broad idea of this project is to be able to implement the following:

- Build an intelligent assistant using principles from Generative AI, RAG, Agentic AI Systems
- Leverage internal enterprise data around frequently asked questions for various departments as depicted below (you are free to modify \ add your own departments - atleast 2 depts for external customers and internal employees should be there). For the purposes of this project you can use LLMs to create the data *(in a real organization scenario you would be sitting with domain experts to create this dataset or even still use LLMs to create it!)*

✅ Departments and Query Types at Company ShopUNow

1. **Human Resources (HR)** – *Internal Employees*
   Handles employee lifecycle matters like leave, payroll, performance, and policy.
2. **IT Support / Tech Support** – *Internal Employees*
   Manages technical issues related to systems, hardware/software, and digital access.
3. **Facilities & Admin** – *Internal Employees*
   Supports employees with workspace logistics, maintenance, access cards, and office supplies.
4. **Billing & Payments** – *External Customers*
   Assists customers with invoice issues, refunds, payment methods, and overcharges.
5. **Shipping & Delivery** – *External Customers*
   Handles customer queries about order tracking, delivery delays, damaged goods, and return pickups.

- Connect the external data to the system leverage principles of RAG
- Leverage patterns like reflection, routing, planning, and \ or multi-agent systems and build the assistant to answer user queries where the user could be an external customer or internal employee

We will show you how to generate sample data you can use to power your Agentic RAG System below. However, you are not compelled to use the sample data we provide for this project, if you have your own documents or data, you are more than welcome to use it (but validate your data and idea with us first), the project steps mentioned below would remain unchanged.

You would anyway need to generate data for some of the departments mentioned above (creating synthetic data has also increasingly become important in the age of Generative and Agentic AI).

Project Goals:

Learning Objectives:

By doing this project it will help you with the following:

- Reinforce your knowledge and skills learnt throughout the program
- Focus on building a project leveraging some of the most common and important aspects of agentic system building in the industry - LLM routing, Agentic RAG, combining multiple agentic workflows

The major project goals or objectives would include the following. You need to do all the key goals mentioned in the compulsory goals section and at least 1 stretch goal (you are welcome to do more).

Compulsory Goals:

- Finalize a list of 4 major departments or more for your company, ShopUNow for which you will be first creating knowledgebases which will be used to answer queries. Remember departments include the following (you should choose at least 2 departments which would interface with internal employees and 2 departments which interface with external customers):

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- Create QA datasets based on potential frequently asked questions for each department, here is an example of how you can use LLMs itself to generate this synthetic data. Generate 10-15 QA pairs per department.

  Be careful about the prompt, remember to put proper descriptions about the department, whether it caters to internal employees or external customers as well as how many QA pairs it should create.

> Act as an expert in running the Human Resources department for Company ShopUNow which is a retail company selling a wide variety of products around clothing, DIY (do it yourself), books, toys.
>
> Create a list of 15 Questions and Answers which could be most frequently asked questions for the human resources department typically by internal employees of the company itself. Remember, the HR deparment handles employee lifecycle matters like leave, payroll, performance, and policy.
>
> Return the response as a list of python dicts with 'doc' and 'category' as keys and values would be the question-answer together as a string and 'HR'

Here is a Python list of 15 dictionaries representing frequently asked HR questions and answers by internal employees at **Company ShopUNow**, each with `'doc'` and `'category'` keys:

python Copy Edit

```python
hr_faqs = [
    {
        "doc": "Q: How do I apply for paid time off (PTO)?\nA: You can apply for PTO through the i
        "category": "HR"
    },
    {
        "doc": "Q: When is the monthly payroll processed?\nA: Payroll is processed on the last wor
        "category": "HR"
    },
    {
        "doc": "Q: How can I access my payslips?\nA: Payslips are available in the HR portal under
        "category": "HR"
    },
    {
        "doc": "Q: What is the process for internal job transfers?\nA: Employees interested in int
        "category": "HR"
    },
    {
        "doc": "Q: Who do I contact for updating my bank account or personal details?\nA: Please u
        "category": "HR"
    },
```

- Create multiple vector databases or a single vector database with appropriate

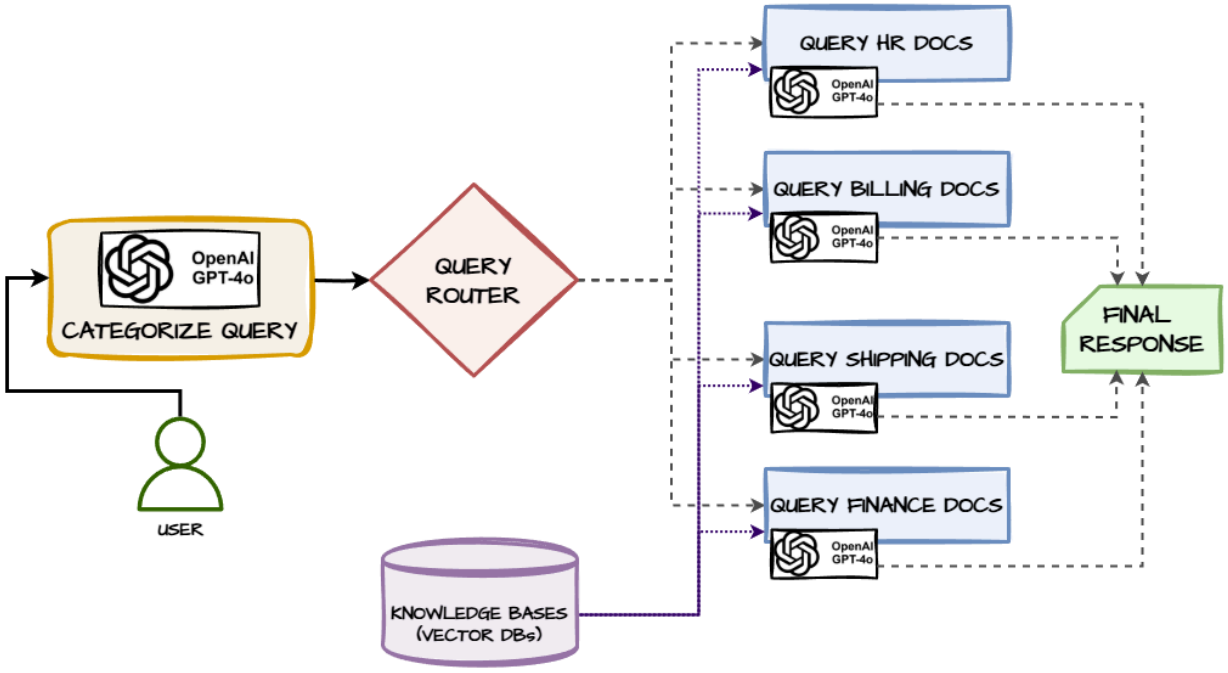
metadata like department which can be used to filter the vector database based on departments when retrieving data

- Build an Agentic AI System which would do the following:
  - ➔ Accept the user query
  - ➔ Figure out the user query sentiment and potential department that could answer the query
  - ➔ If the sentiment is negative OR department is not one among the list of departments, route it to an AI workflow (steps or sub-agent) which would send the query to a human support agent (here you can just return a simple message saying the human agent will call you back)
  - ➔ If the sentiment is positive or neutral, based on the department, route it to a relevant AI workflow (steps or sub-agent) which can use RAG and generate an appropriate response to the user query
- Test the Agentic AI System on sample queries

NOTE: *You are free to use any framework of your choice to build the Agentic AI System using LangChain, LangGraph, CrewAI, AutoGen or anything else.*

STANDARD WORKFLOW:

The major workflow of a Router-based Agentic RAG System is as follows (check out our Agentic AI courses on LangGraph to go through a guided project on Router Agentic RAG Systems if you need more deep dives). You can use a similar architecture or any other of your choice as long as it achieves the above goals



Stretch Goals (at least 1):

- Advanced Option 1: Enhance this system into a multi-user conversational Agentic system and manage memory appropriately.
- Advanced Option 2: Add more interactivity to the human escalation route, add in a form that takes in input from the user - basic details like name, number, email and then shows the info back to the user saying this info has been received, someone will reach out to you soon. You can even connect email or WhatsApp APIs to simulate sending an email or message
- Advanced Option 3: Add more routes to the agent and make it more diverse including more departments and any custom workflows of your choice
- Advanced Option 4: Wrap the agent in an API or application using tools like FastAPI, Flask etc, and then deploy the web service and test it out with sample queries

Milestones:

- Finalize your tool stack - LangChain, LangGraph, AutoGen, CrewAI, Python, No-code - you are free to choose
- Create Datasets for VectorDB as mentioned above
- Create your Agentic AI System
- Build an enhanced Agentic AI system based on stretch goals
- Test your enhanced Agentic AI system

Evaluation:

You are recommended to meet with a mentor to review on your progress and then do a final project walkthrough and demo of your project once you are ready with a presentation and then submit it.